Technical Documentation

ACDT Project 1, GROUP32: Is daily social media use associated with student mental health?
Website: dhkim060225-web.github.io/group32-sns-mental-health
Code: github.com/dhkim060225-web/group32-sns-mental-health

1. Research question and hypotheses

Outcome Y = Mental_Health_Score; treatment X = Avg_Daily_Usage_Hours. Model: $Y = \beta_0 + \beta_1 X + \varepsilon$. H_0: $\beta_1 = 0$; H_1: $\beta_1 \neq 0$ (two-sided; our earlier directional version was $\beta_1 < 0$, controlling for academic level). **$\alpha = 0.05$**: a Type I error would justify blanket screen-time rules and wasted counselling time; a Type II error would leave struggling heavy users unflagged. Neither cost dominates, so we kept the conventional level.

2. Data and preprocessing

Source: *Students' Social Media Addiction* (Kaggle), 705 rows, 13 self-reported variables, no missing values. Recruitment method and sampling frame are not documented (treated as a convenience sample). Student IDs 242 and 274 are identical in every answer; one was removed (**n = 704**). 110 countries, median 1 student per country. For DML, Country was recoded to the 10 largest countries plus "Other".

Variable	Type / range	Role	Reason
Avg_Daily_Usage_Hours	Continuous, 1.5–8.5 h	Treatment (X)	The conventional wisdom is about time spent
Mental_Health_Score	Integer, observed 4–9 (1–10 scale)	Outcome (Y)	Replaces GPA, which is not in the data
Age, Gender, Academic_Level, Relationship_Status	Background	Control	Fixed before usage; can confound, not caused by usage
Country	110 categories	Control / cluster	Strongest confounder; breaks independence
Most_Used_Platform	12 categories	Moderator	Proxy for type of use (our original idea)
Sleep_Hours_Per_Night	Continuous	Mediator	Usage can reduce sleep; not controlled for the total effect
Addicted_Score, Conflicts_Over_Social_Media	Scores / counts	Not controlled	Likely consequences of usage (bad controls)
Affects_Academic_Performance	Yes / No	Secondary outcome	Closest proxy for the original GPA question

3. Methods

Framework level	Method	Implementation details
Sampling distribution / Inference	OLS, t-test, 95% CI	df = 702, $t^* = 1.963$. Independence violated by country clustering, so CR1 cluster-robust SEs with 110 clusters are also reported. Residual normality: Shapiro–Wilk $p < 0.001$; with n = 704 the CLT supports t-based inference.
Prediction	Train/test validation	200 random 80/20 splits (scikit-learn, seeds 0–199); metrics: test R^2, RMSE, vs a train-mean baseline. Prediction intervals: $\hat{y} \pm t^* \cdot s\sqrt{(1 + 1/n + (x-\bar{x})^2/S_{xx})}$.
Inference (heterogeneity)	Per-platform OLS; nested F-test	Bootstrap CIs (2,000 resamples) for slope and R^2. F-test: platform-specific slopes vs one common slope, both with platform intercepts (platforms with $n \geq 10$ and varying outcome).
Reasoning	Country fixed effects; mediation	110 country dummies. Sleep mediation by product of coefficients a·b, 1,000 bootstrap resamples.
Reasoning	Double/debiased ML (partially linear model)	$Y = \theta X + g(W) + \varepsilon$, $X = m(W) + v$. Random forests (100 trees, min leaf 5; 8 in subgroups) estimate g and m with 5-fold cross-fitting; θ from residual-on-residual regression; 3 repetitions, median aggregation; influence-function SE. Covariate sets A–D below.

4. Results

Model	Estimate per hour	95% CI	R^2
OLS: hours only	−0.704	−0.74 to −0.66 (clustered: −0.80 to −0.61)	0.642
OLS: + academic level	−0.691	−0.73 to −0.65	0.657
OLS: + sleep (direct effect)	−0.566	−0.63 to −0.50	0.657
OLS: + country fixed effects	−0.575	−0.62 to −0.53	0.848
OLS: + country + platform	−0.521	−0.57 to −0.47	0.874
OLS: + sleep + addiction + conflicts	−0.063	−0.11 to −0.02	0.896
DML A: background (age, gender, level, relationship, country)	−0.633	−0.68 to −0.59	–
DML B: A + platform	−0.589	−0.65 to −0.53	–
DML C: A + sleep	−0.454	−0.52 to −0.38	–
DML D: A + addiction + conflicts	0.056	0.01 to 0.10	–

Validation: mean test R^2 0.635 (range 0.484–0.730), RMSE 0.662 vs baseline 1.104. Platform slope heterogeneity: $F(8, 650) = 3.59$, $p = 0.0004$; academic level $p = 0.81$; gender $p = 0.31$. Mediation: $a = -0.709$, $b = 0.194$, indirect = −0.138 (95% CI −0.177 to −0.104), 19.6% of total. DML subgroups: Undergraduate −0.63, Graduate −0.68, High School −0.70, Female −0.63, Male −0.69, Instagram −0.66, TikTok −0.54, Facebook −0.66; WhatsApp could not be estimated because scores are constant within its countries. Secondary outcome: students reporting academic harm use 5.54 vs 3.80 h (Welch $t = 25.4$, $p < 0.001$).

Interpretation of D: controlling for addiction and conflicts removes the effect because both are probably downstream of usage and strongly tied to the outcome ($r = -0.95$ for addiction). We report D only to show why such variables must not be controlled.

5. Limitations and threats to validity

- **External validity:** undocumented, non-random sample; half of the 110 countries contribute one student.
- **Internal validity:** cross-sectional self-reports; reverse causation and unmeasured confounding (stress, workload, purpose of use) remain. DML adjusts only for measured covariates; no valid instrument was available.
- **Data quality:** scores identical for all users of LINE, KakaoTalk and VKontakte and within 13% of countries with $n \geq 2$; academic-impact answers move from 0% to 100% across usage bands; outcome is ordinal with six observed values.
- **Model assumptions:** non-normal residuals and clustered observations; addressed with large-sample inference and cluster-robust SEs, not eliminated.

6. Reproducibility

Repository files: **analysis.py** (all computations, writes results.json), **results.json**, **data/Students_Social_Media_Addiction.csv**, **index.html** (data story website), **tools/** (page and document builders). Environment: Python 3 with pandas, numpy, scipy, scikit-learn. Run **python analysis.py** (about 1–2 minutes), then **python tools/build_page.py** and **python tools/build_docs.py**. All random procedures use fixed seeds.

AI assistance: *An AI assistant (Claude, by Anthropic) was used to draft the analysis code, the web page, and the two PDF documents. The team defined the research question, chose the variables and methods, and reviewed all interpretations. Every reported number was checked by re-running analysis.py on the course dataset.*